



Daesung Experiments TiO_x on SiC



10 September 2026

CONFIDENTIAL 2026

Parameters of Coating experiments

Test dates of data shown:

01-07-2026 till 02-07-2026

Powder:

- Coating of TiOx on SiC
- SiC Powder Ti content without coating: <0.005wt%

Parameters:

- Temperature of the reactor: 200 °C
- Temperature of the TTIP Bubbler/CEM: 100 °C
- Temperature of the Water Bubbler: 40 °C
- Batch Size: ~1.6 - 1.8 kg
- Expected throughput: 2.5 & 5 kg/hr
- For each test new batch of powder was used.



Overview of Tests

Test	Target	Description	Expected throughput kg/hr	System used
A	Create thick coating in a few cycles by using near-saturation conditions	High precursor dose, low powder throughput	2.5	A (CEM)+C (Bubbler) (A vapour + C gaseous)
B	Test if dosed chemicals are all deposited on particles	Low precursor dose, high powder throughput	5	A (CEM) +C (Bubbler) (A vapour + C gaseous)
C	Test if bubbler and CEM system work identical at low dose conditions	Bubblers only: Low precursor dose, high powder throughput to compare with B	5	B (Bubbler) +C (Bubbler) (B gaseous + C gaseous)

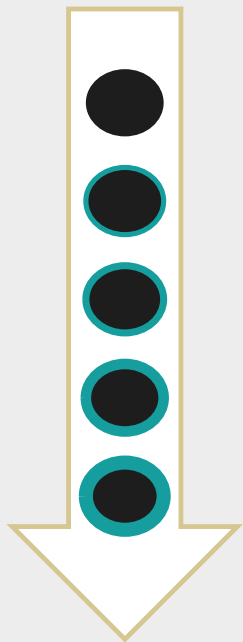


Test A, High precursor dose: Experimental Parameters

Setup: 2 Precursor system (A+C) (vapour + gaseous), TTIP & water on SiC

Expected throughput: 2,5 kg/hr

The same batch of powder was used for sequential coating in this test



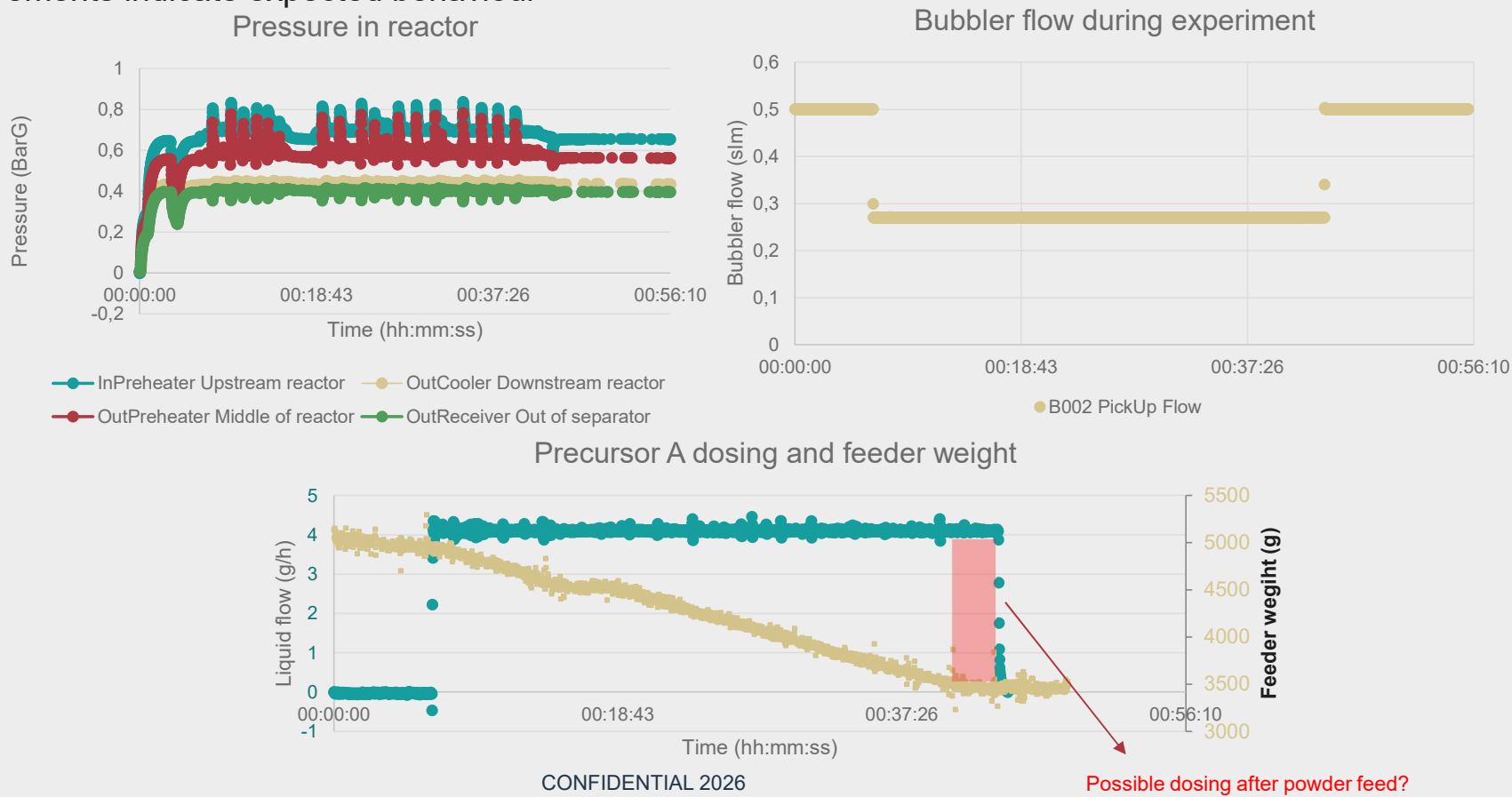
Description	Injection Points used	Accumulated no. of cycles	Liquid flow precursor A (g/h per injection point)	Flow through the CEM MFC (slm per injection point)	Gas flow precursor C (slpm)
POS03- 003	1	1	4,1	2	0,27
POS03-004	1	2	4,1	2	0,27
POS03-005	2	4	4,1	2	0,27
POS03-006	2	6	4,1	2	0,27



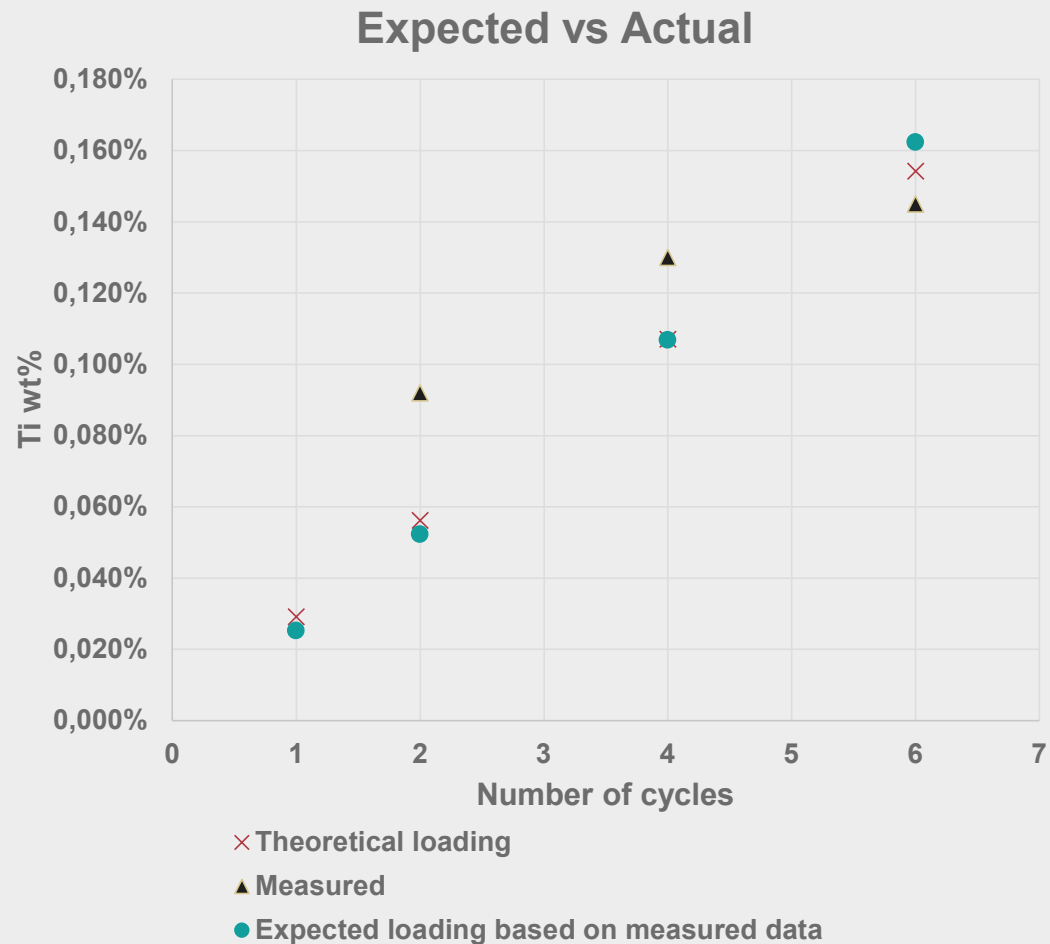
Measurements during experiment A, sample POS03-004

- Main process parameters shown
- In this dataset weight controller stopped temporarily around 00:15:00
- All measurements indicate expected behaviour

Description	Through put (Actual) (kg/hr)
POS03- 003	2,4
POS03-004	2,5
POS03-005	2,7
POS03-006	2,9



TiOx loading results test A



- High difference between measured and theoretical loading in cycle 2:
- Possible cause is a sampling error: e.g. collected unreacted Ti from backpulse filter.
- Carry over from previous experiment is unlikely: previous run had 0.018% loading, and system was cleaned before experiment.
- Recalculation shows 3% underdosing of H₂O was happening during the experiment. Such undersaturation may be cause of lower growth per cycle from 2 → 4 and 4 → 6.
- The reason for initial double amount of loading unresolved (0.04% → 0,72 g of TiO₂).

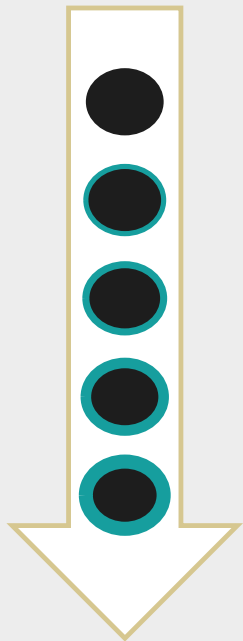


Test B, low precursor dose: Experimental Parameters

Setup: 2 Precursors (A+C) (vapour + gaseous), TTIP & water on SiC

Expected throughput: 5 kg/hr

The same batch of powder was used for sequential coating in this test.

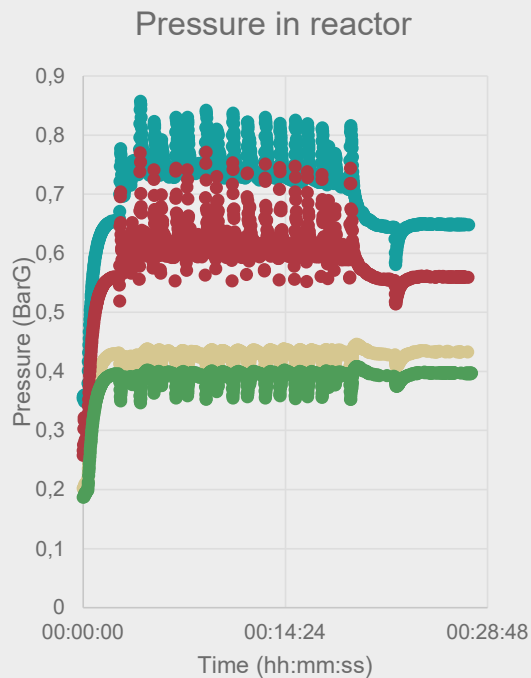


Description	Injection Points used	Accumulated no. of cycles	Liquid flow precursor A (g/h per injection point)	Flow through the CEM MFC (slm per injection point)	Gas flow precursor C (slpm per injection point)
POS04-001	1	1	1,04	0,52	0,06
POS04-002	1	2	1,04	0,52	0,06
POS04-003	2	4	1,04	0,52	0,06
POS04-004	2	6	1,04	0,52	0,06

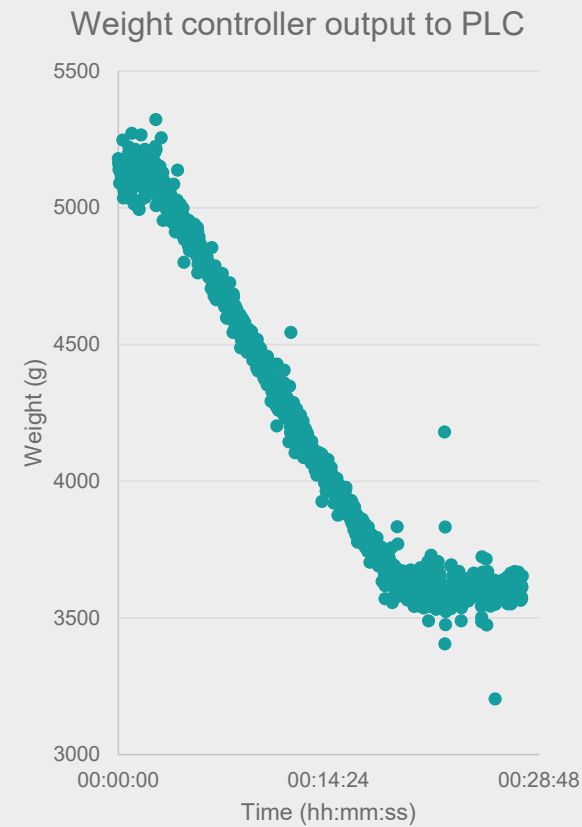
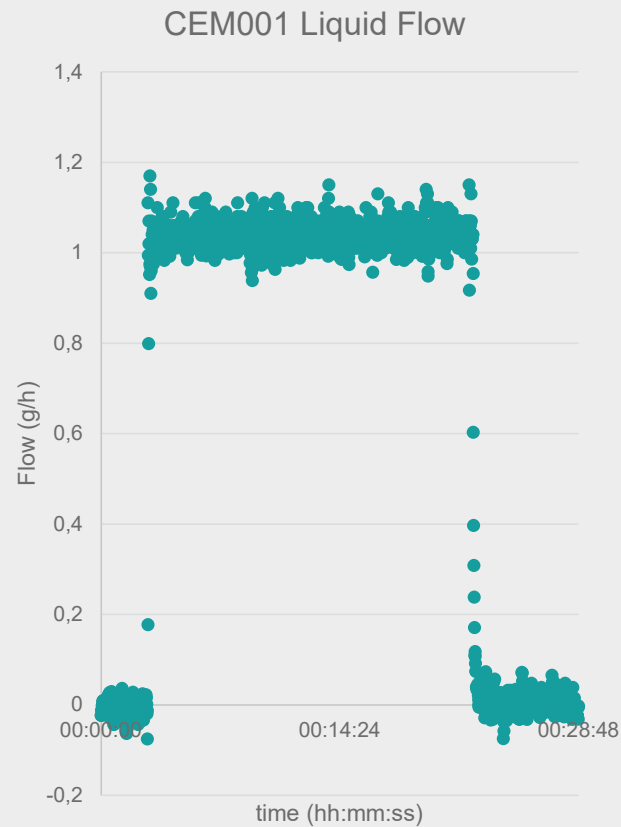


Measurements during experiment B, sample POS04-001

- All operational measurements indicate expected output.

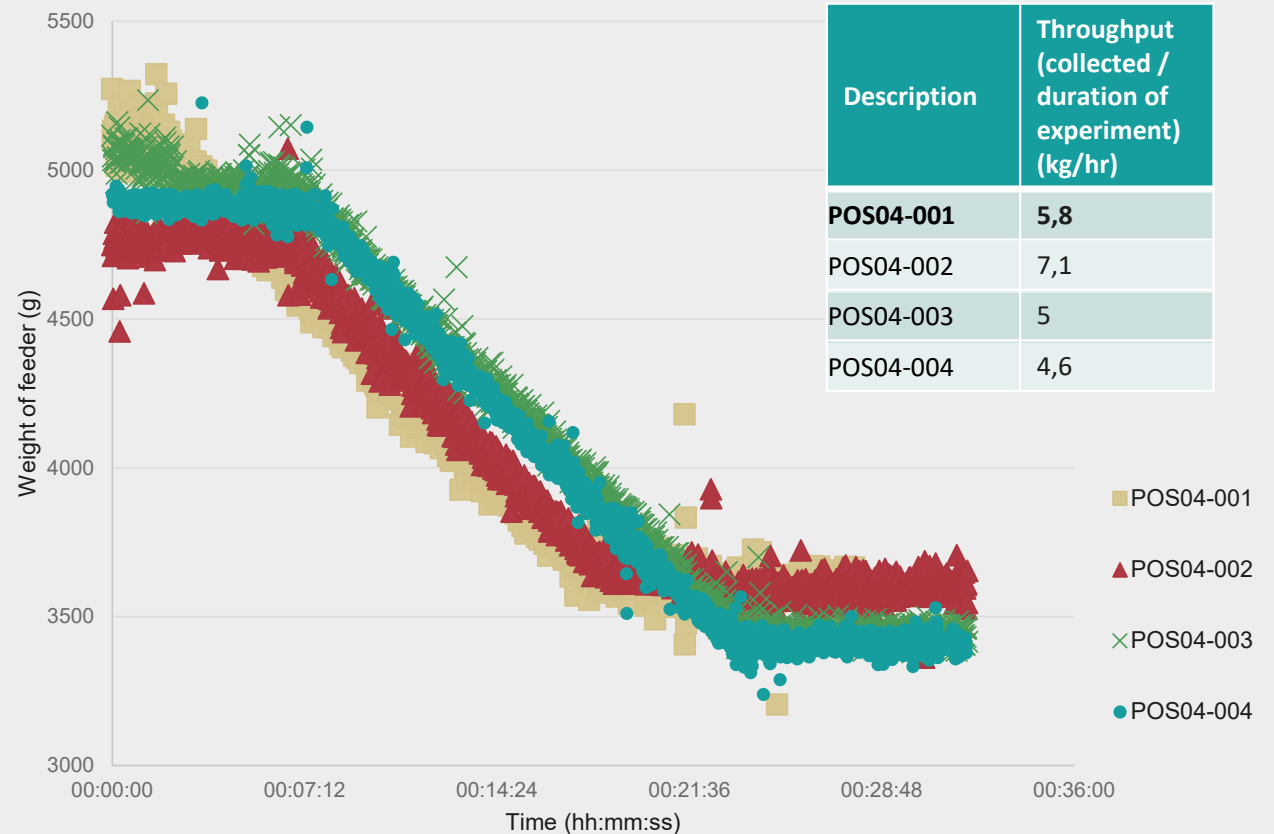


● InPreheater Upstream reactor
● OutCooler Downstream reactor
● OutPreheater Middle of reactor
● OutReceiver Out of separator

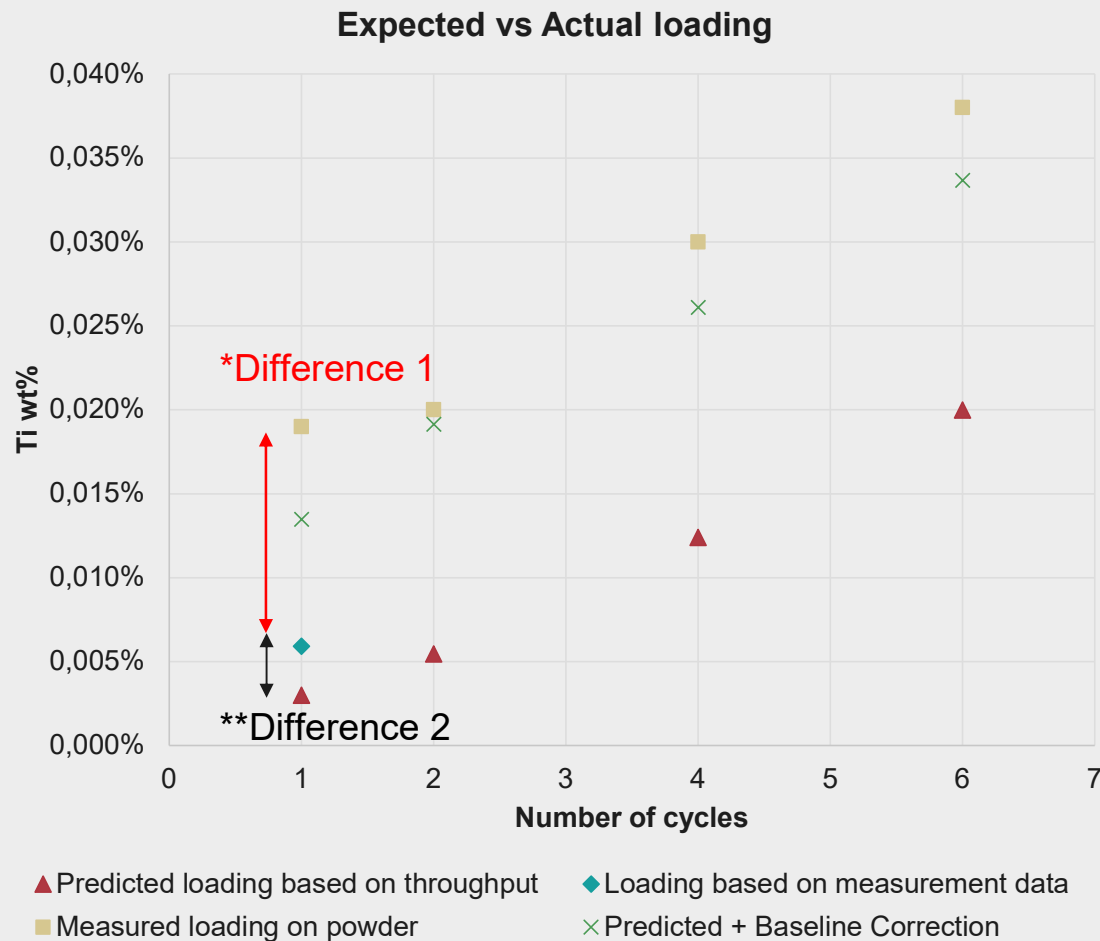


4 experiment comparison of powder dosing in WIC with static setpoint

- The same linear decrease in weight over time indicates the actual powder flow was very steady during every experiment.
- The equipment environment was suffers from vibration, causing low measurement precision.
- The throughput based on the collected powder amount deviates, indicating some powder was not collected.



TiOx loading results Test B



Achievements:

- Ti loading shows linear increase after 1st cycle.
- Ti growth in line with predicted growth

*Difference 1

- Either there is Ti coming from backpulse filter (similar as experiment A),
- Or 9% carry over from previous experiment,
- Or unreacted precursor in system from previous experiment (0.0105 w% Ti total),
- Or a combination of the above.

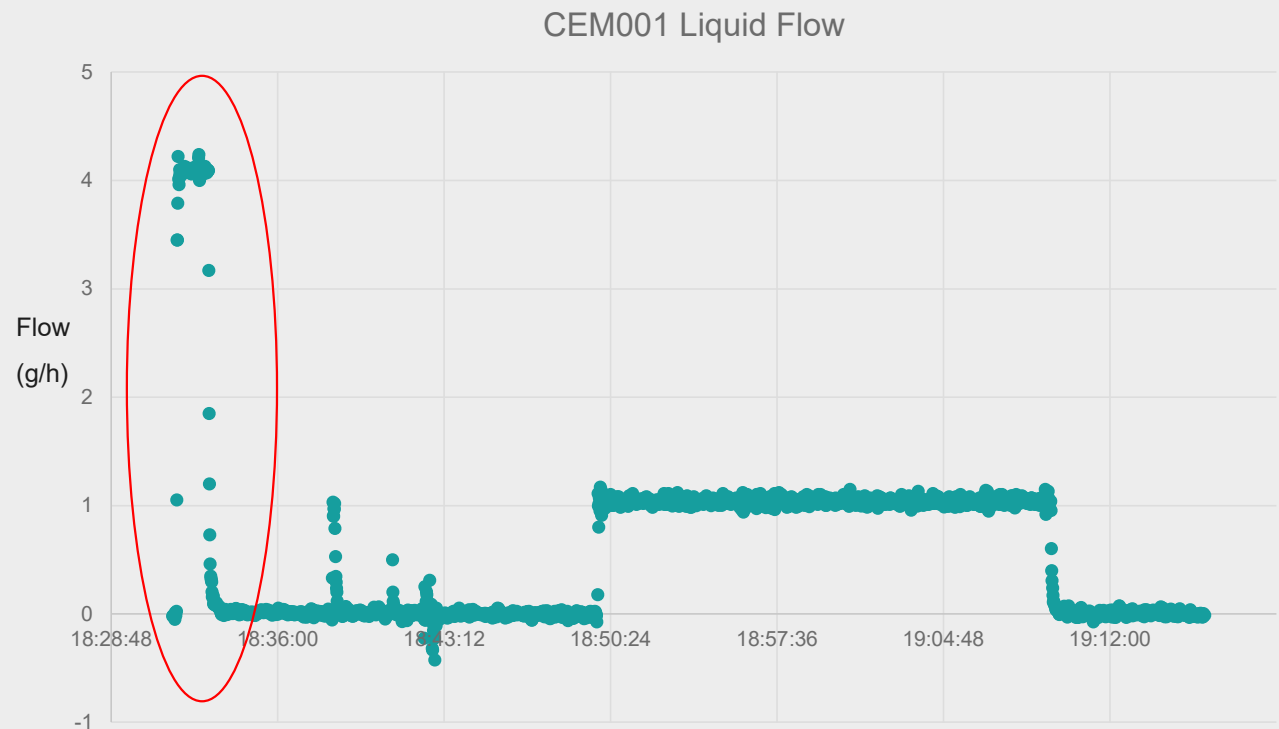
** Difference 2: shown in next slide

Baseline correction: 0.0137% Ti base loading at cycle 0)



**Overdosing precursor 1 during first cycle of test B

- During the onset of starting test B the setpoint flow was not sent/picked up by PLC.
- As a result, a dosing period with the old setpoint was used.
- This additional TTIP precursor amount was added during the first cycle only.

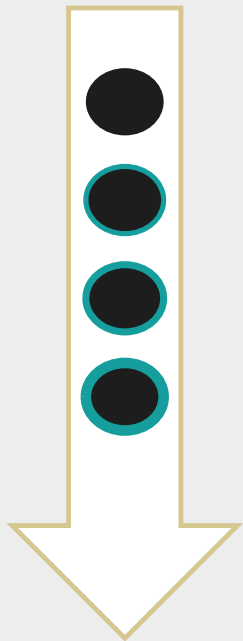


Test C, bubblers only: Experimental Parameters

Setup: 2 Precursor system (B+C) (gaseous + gaseous), TTIP & water on SiC

Expected throughput: 5 kg/hr

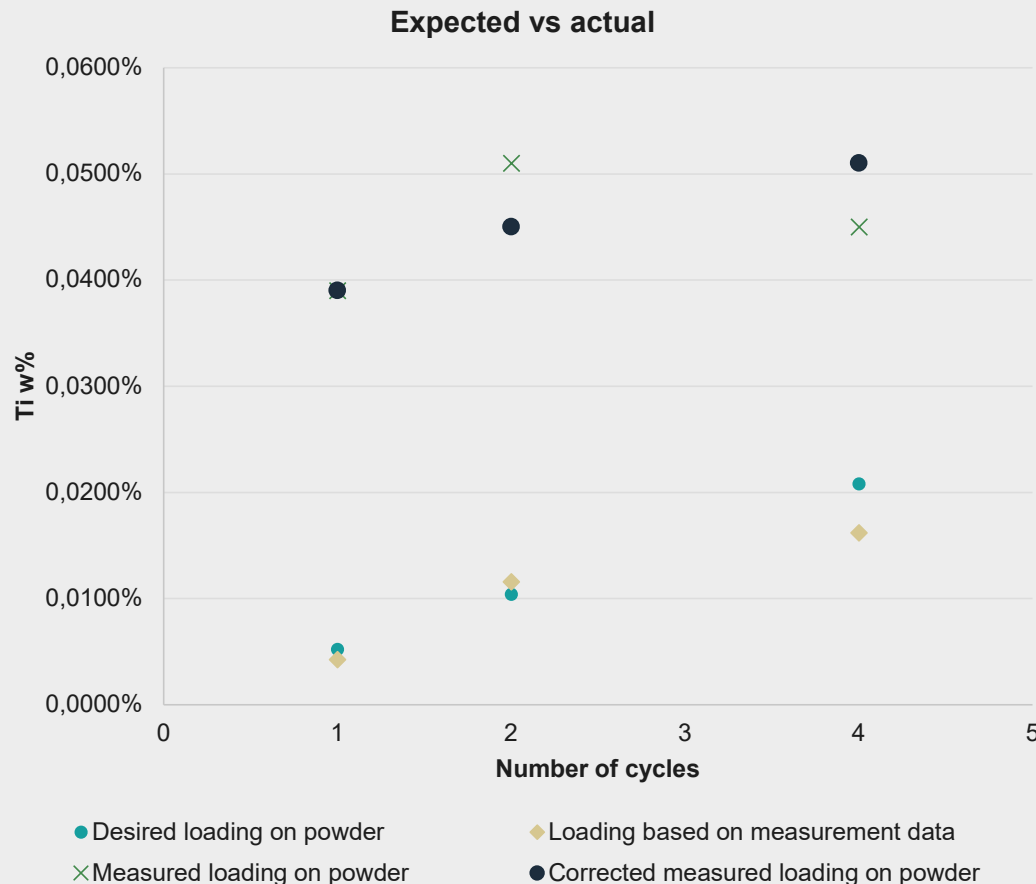
The same batch of powder was used for sequential coating in this test.



Description	Injection Points used	Accumulated no. of cycles	Gas flow precursor B (slpm)	Gas flow precursor C (slpm per injection point)
POS05-001	1	1	0,52	0,06
POS05-002	1	2	0,52	0,06
POS05-003	2	4	0,52	0,06



TiOx loading results Test C



Again, a very high base loading is found.

- In this case it would amount to 100% fraction of previous batch. This is not possible.
- More probable is TiOx contamination from a previous run, e.g. this could have come from coated powder stuck at the filter
- Another reason can be that the powder from test B is used (which would be in line with the expected loading).

The samples 2 and 4 appear to be swapped. If they are interchanged the expected growth is observed (Corrected measured loading on powder)



Part two: comparing test A, B and C

To understand the behaviour of the machine, a combinatorial analysis has been carried out.

A and B: similar parameters except the throughput (1) and the dose (2)

- Unknown if saturation condition was there during test A.
- Unknown effect of variability of measured powder throughput (see graph on the right):
- Expected: Ti growth during test A is a multiple of test B.

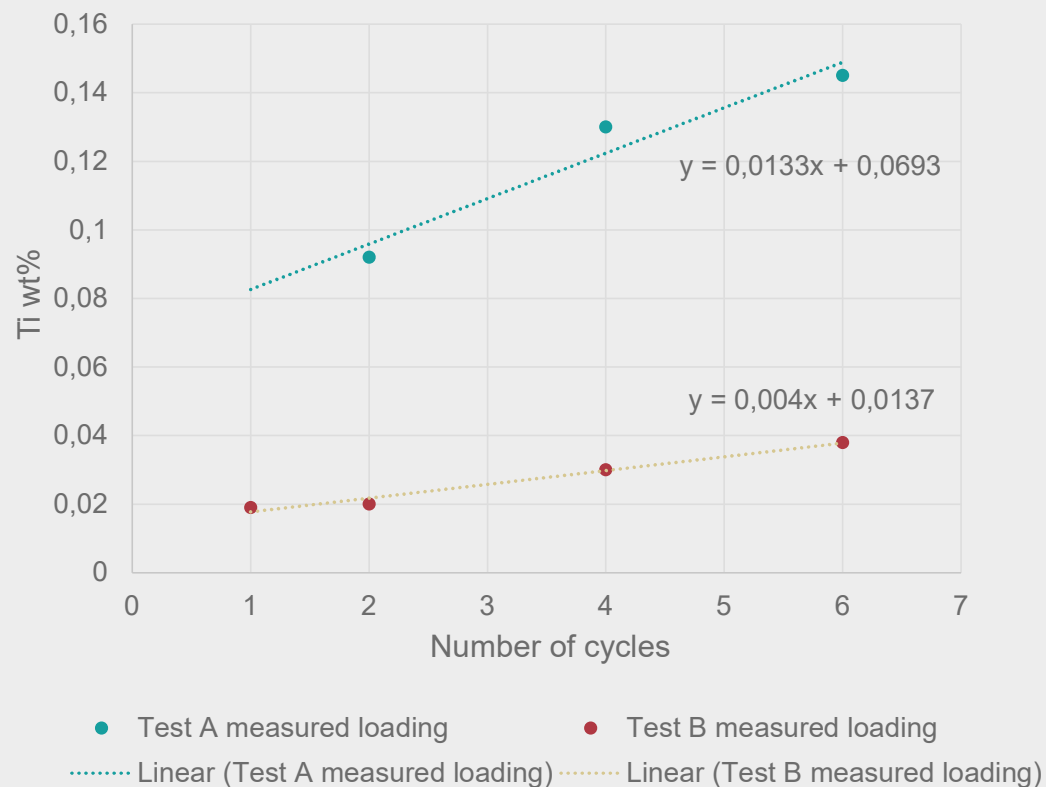
B and C: same parameters using other dosing method of precursor A

- Same Ti% dose targeted during experiments
- Same powder throughput used



Test A vs Test B: more precursor, more coating

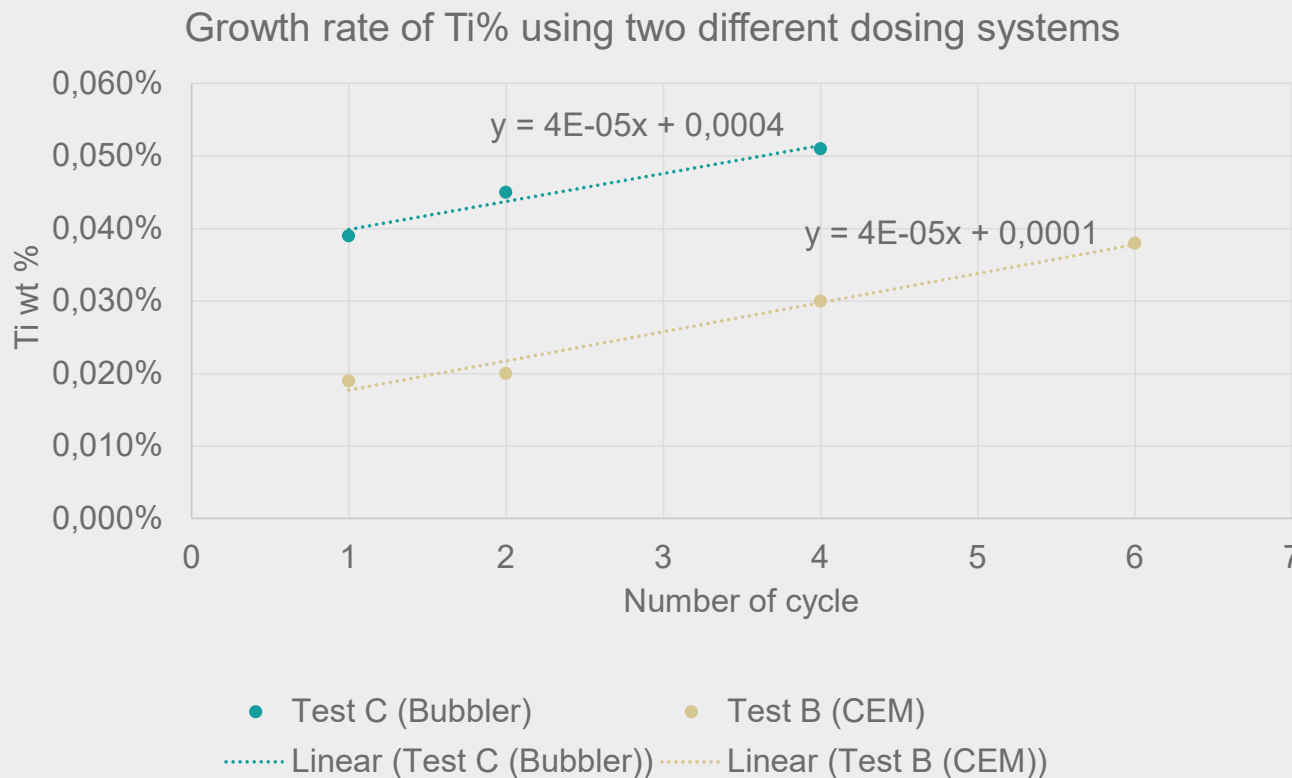
Growth rate of Ti% on SiC, using two dosing conditions



- Test A doses about 8 times more Ti (4.1 vs 1.0 g/h with 2.5 kg/h instead of 5 kg/h powder flow).
- However, we find a multiplication of 3.3 times in weight gain of Ti.
- Difference may be caused:
- **Saturation** during high dose exp (dose is based on BET fully saturated without FBR tests).
- Underdosing water in high dose experiment. This amounts to 3% → would show 3.9 times instead of 8 times the weight gain.



Test B versus Test C: do we get the same growth?



- Ti% loading versus cycles from test B and C shown (C sample numbers 2 and 4 are swapped).

- **Dosing response of using CEM and bubbler as precursor A source shows same growth (slope in the graph)**



Conclusions, part 1

The tests show the expected Ti growth during the experiment:

1. Test A produced the highest measured Ti loading. Oversaturation of TTIP is a probable cause of lower growth per ALD cycle, setting the boundary of the allowable Ti growth.
2. Test B shows the clearest linear increase in Ti loading as function of ALD cycles.
3. Test B and C show same ALD growth per cycle, which confirms dosing via CEM and bubbler are equivalent methods. The bubbler has physical limitations at higher dosing rates, making the CEM a superior method for some coatings.



Conclusions, part 2

All three tests show higher reported Ti loading than expected. There are multiple suspect causes:

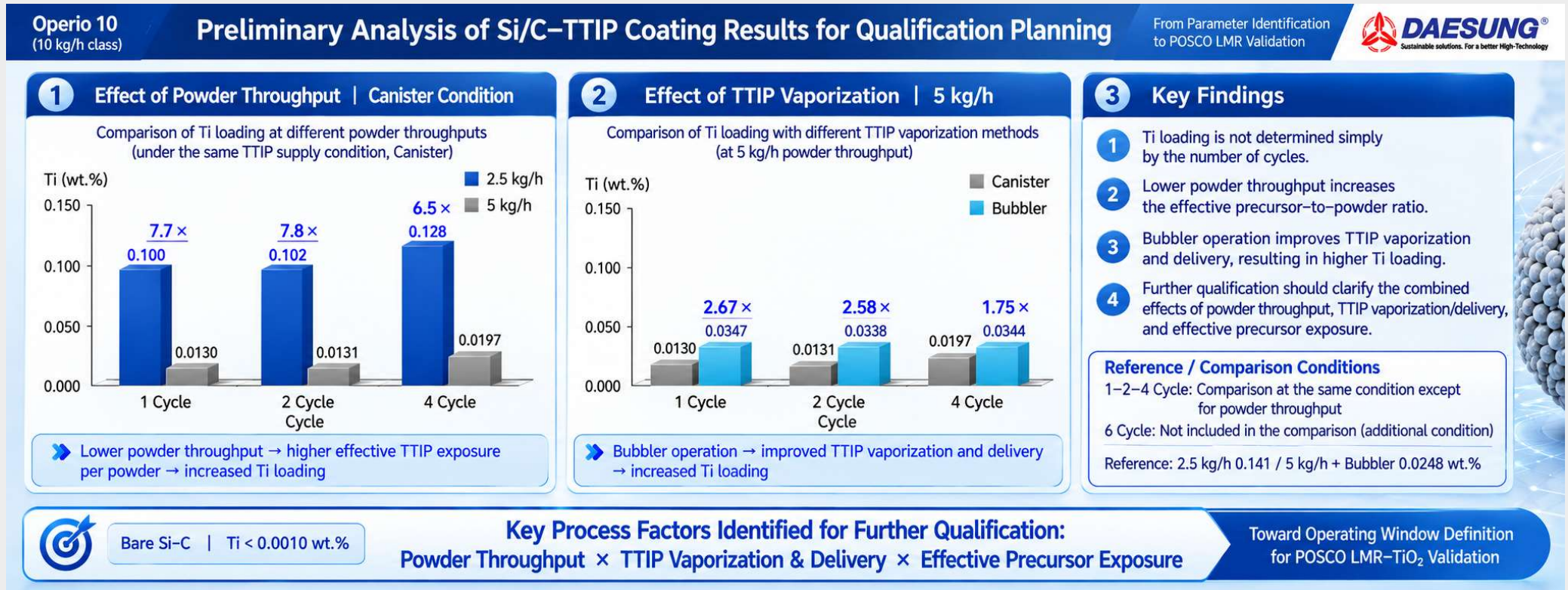
1. For Test A the only probable cause is buildup of Ti in the backpulse filter after dosing Ti in an empty reactor before.
2. For Test B the higher loading is probably partially originating from powder from the previous experiment.
3. For Test C probably there is a bag swap, using test B material: the starting material appears to have the same loading as the end result of test B. If this is not the cause, the problem of Ti residuals getting stuck in the backpulse filter is major.

Lessons learned:

- More thought should be given to the sampling procedure: to rule out variability and possible sources of contamination.
- Sampling before and after conducting backpulsing should be tested to find out if the higher loading originates from residual reactants in the filter.
- Additional backpulse filter cleaning between experiments should be carried out to find out if carryover between experiments causes a higher loading.



Comparison of analysis Daesung

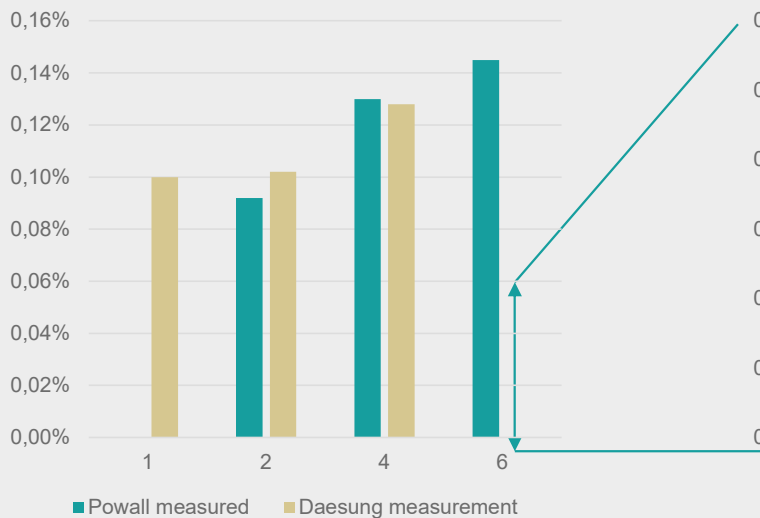


Preliminary results sent 10-09-2026
by Daesung



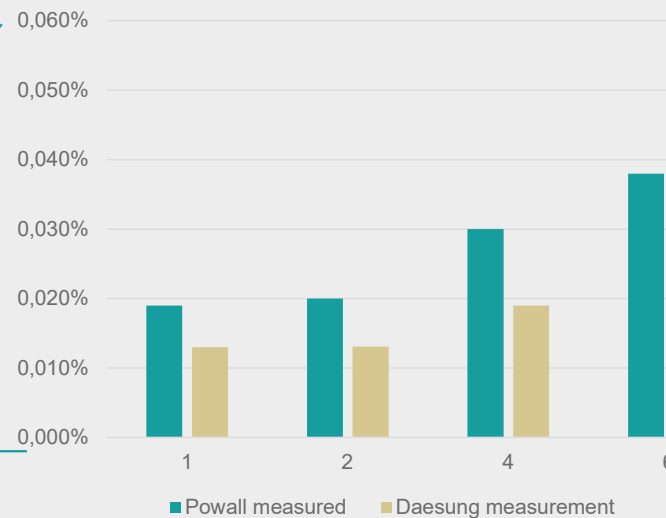
Measurement differences in sample Ti content

- Samples were taken at the same time at the same place by the same operator.
- A high deviation between the two analytic laboratories observed. This might indicate powder mixture was inhomogeneous.
- Less of expected Ti growth trend found in Daesung data. Order of value is similar.



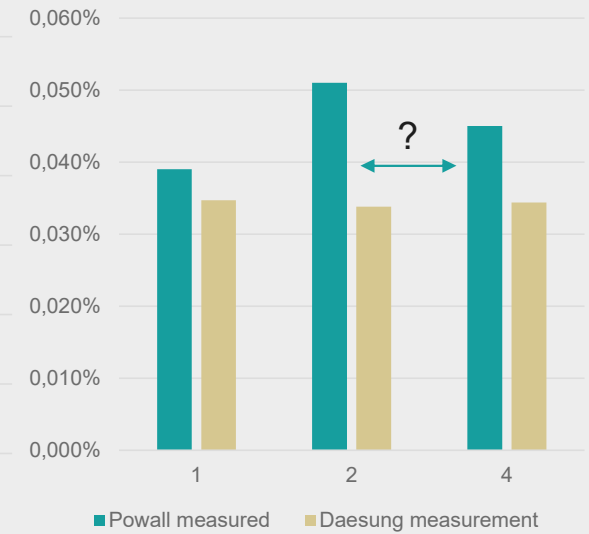
High dose, low throughput

Test A



Low dose, high throughput

Test B



Low dose, high throughput, bubbler

Test C



Interpretation of findings

3 Key Findings

- 1 Ti loading is not determined simply by the number of cycles.
- 2 Lower powder throughput increases the effective precursor-to-powder ratio.
- 3 Bubbler operation improves TTIP vaporization and delivery, resulting in higher Ti loading.
- 4 Further qualification should clarify the combined effects of powder throughput, TTIP vaporization/delivery, and effective precursor exposure.

Reference / Comparison Conditions

1–2–4 Cycle: Comparison at the same condition except for powder throughput

6 Cycle: Not included in the comparison (additional condition)

Reference: 2.5 kg/h 0.141 / 5 kg/h + Bubbler 0.0248 wt.%

1. During these tests the Ti content measured indeed is not determined by the number of cycles only. Given ALD, the coating of Ti is determined by the number of cycles. However, there appears to be contamination in the powder.
2. During the tests with high and low powder throughput, a net 8 times difference in loading was targeted. We therefore cannot state if the precursor deposition is affected by the powder throughput from these experiments. This was not the target of these experiments.
3. We see from cycle 1 → cycle 2 → cycle 4 a similar increase in Ti loading between bubbler and vaporization method. The base powder seems to have contamination.
4. The conducted trial run has shown that presence of residual powder and reacted precursor is affecting the measured Ti content more than the qualification parameters.



Looking forward

- Additional deposition tests using various precursor dosing rates for multiple cycles are necessary to validate the accuracy and precision of the system.
- Before starting such trials however, it is important to find out where the Ti contamination is coming from. During the previous trials the Ti content often exceeded the amount of Ti dosed in the complete experiment.





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